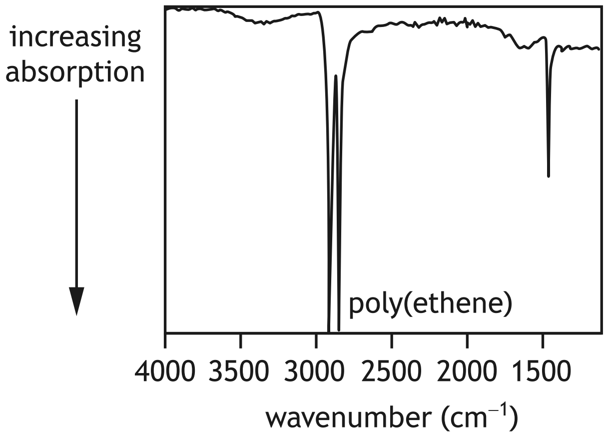
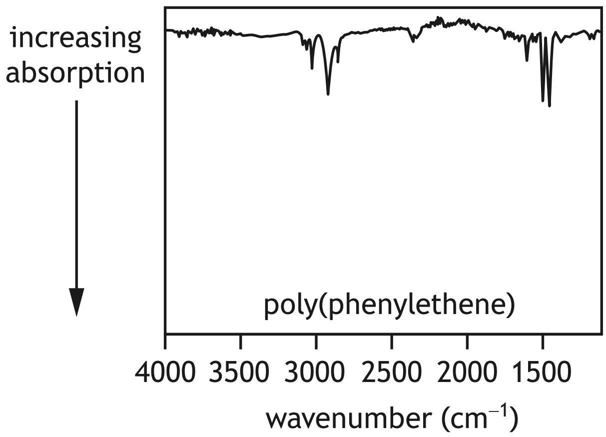
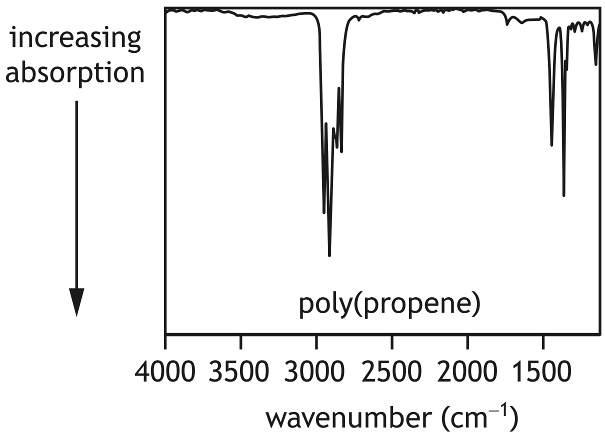
12. One of the problems with recycling plastics is identifying the type of plastic.

Infrared spectroscopy is a technique that can be used to identify the bonds present in plastics. A spectrum is produced for each sample analysed. The same bond always absorbs infrared radiation in the same range of wavenumbers, even in different molecules. For example C-H bonds absorb in the wavenumber range 2700–3300 cm^{-1}.

Four different types of plastic were analysed using infrared spectroscopy and the spectra produced are shown.







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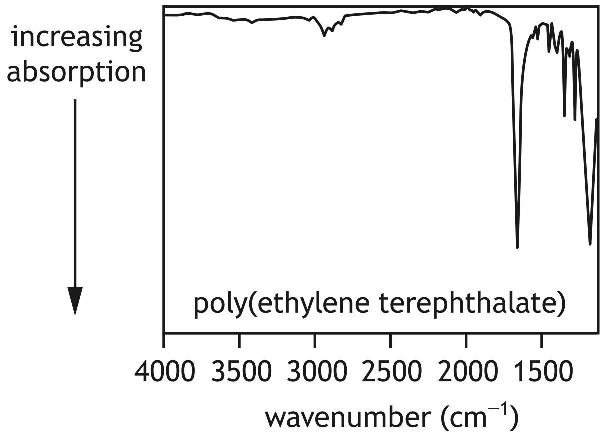
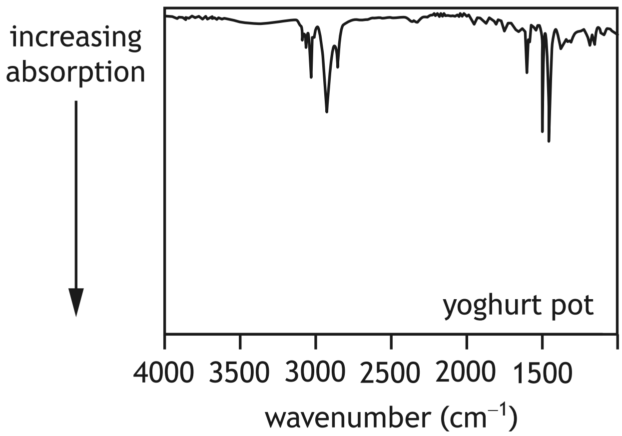
12. (continued)



(a) The spectrum obtained from the analysis of the plastic used to make a yoghurt pot is shown.



Identify the type of plastic used to make the yoghurt pot. **1**

[Turn over

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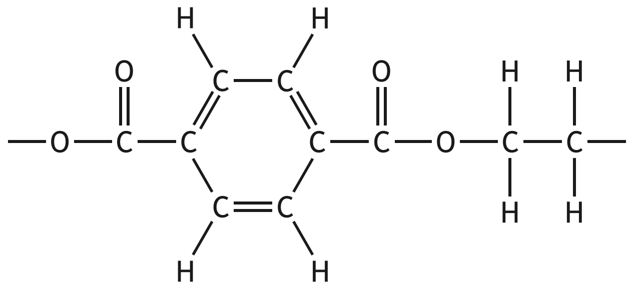
12. (continued)

(b) The spectrum produced from poly(ethylene terephthalate) contains an absorption at a wavenumber of 1720 cm^{-1}.

Part of the structure of poly(ethylene terephthalate) is shown.



Using the information on page 14 of the data booklet, circle the bond in poly(ethylene terephthalate) that is responsible for this absorption. **1**

(An additional diagram, if required, can be found on *page 40*.)